# TESTED AND HOPED ON

control software for the physical world, and the language that proves it

Nobody can prove a controller will never reach the network while it is steering a seven-ton flywheel. It is tested, reviewed, and hoped on.

Agents now write that software. The grid is where it bites first.

### WHY NOW

Coding agents write production control code this year.

### WHY THE GRID

Every new inverter is a software product under UL 1741 SB, and Utah's data centers are the load.

### WHY HERE

Torus is scaling a factory in Salt Lake; Hill AFB verifies flight software by review.

# MENTL

a language whose compiler proves what a program can never do

## NEVER

"This function never reaches the network, never allocates, never exceeds this bound" is a claim the compiler proves through every call, or the program does not build.

## WHY

Every fact carries its reason, walkable to the line that decided it. A dispatch explains itself.

## LEARNS

A learning rule is read off the chain it learns on, no tape, no second copy to drift.

### THE SAME PROGRAM, ONE LINE ADDED

REFUSED

```
effect Net {
  fetch(url: String) -> String
}

fn decide(reading, station) with !Alloc + !Network + !Filesystem = {
  let price = fetch("https://api.eia.gov/price")   // scale the slow band by price
  ...
}
```

`decide` → `fetch` → `Net` the path the network takes into a function that swore it never would

```
E_EffectMismatch: effect row mismatch: !Network vs ... Net — declared as decide; Net enters through `fetch`, an op of Net.
```

This is what review cannot do: the build refuses, and names the line.

Self-hosting: the pinned compiler reproduces itself byte for byte. One WebAssembly file. Open source. [[years]] of commits.

# BUILT THIS WEEK

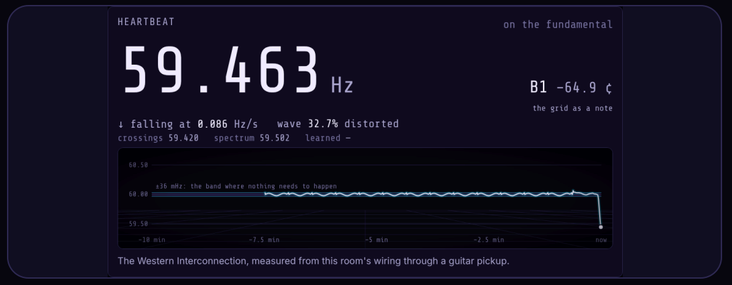
beside what existed before

## BEFORE KICKOFF

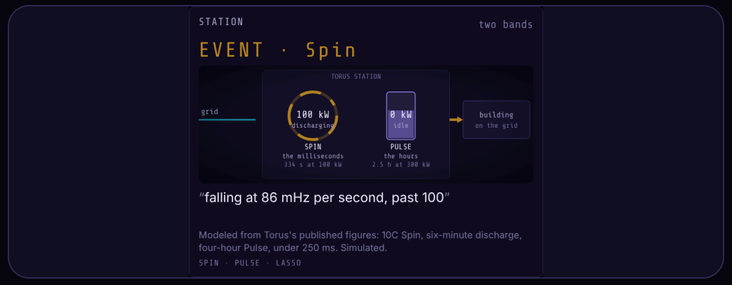
- Mentl: the compiler, the proofs, the course, the Wasatch piece made of Salt Lake's readings
- sonido: 36 effects, the plugin, the browser editor, the pedal firmware

## SINCE MONDAY 4 PM

- The ear: grid frequency from a guitar, three ways, to the millihertz
- The controller in Mentl, its proofs, its refusal twin, the learner
- The page, the director, the data, the real event replay
- The pedal as a second ear; an agent's proposal refused by the gate



The Western Interconnection, measured from this room's wiring through a guitar pickup.



"falling at 86 mHz per second, past 100"

Modeled from Torus's published figures: 10C Spin, six-minute discharge, four-hour Pulse, under 250 ms. Simulated.

SPIN · PULSE · LASSO






./LEDGER.SH PRINTS EVERY COMMIT WITH ITS TIME

# WHO NEEDS IT, IN UTAH

### TORUS

Their Station hears every cycle. Their security is a perimeter. The program itself has never been provable, until now. The Torus judge heard this demo today.

### HILL AFB, 309TH SOFTWARE ENGINEERING GROUP

Utah's largest employer of software developers, verifying flight software by review. A proof is a deliverable they can file.

### EVERY UTAH COMPANY PUTTING AGENTS ON PRODUCTION CODE

A gate that answers PROVEN or REFUSED, with the line, before anything an agent wrote runs.

First hires: compiler and controls engineers, Salt Lake City.

# WHAT $10,000 UNLOCKS

ninety days, two introductions, one co-founder

### Day 30

The proven controller on inverter-class hardware on my bench. SAM.gov and SBIR.gov registered. Company formed. [[date]]

### Day 60

One Utah site, one ear on a real Station-class inverter, the dispatch explained to its operator. The SBIR Phase I draft with Nucleus Grow.

### Day 90

Ninety days of decisions on the page, the proposal filed, the first paid pilot signed.

The business: paid pilots with asset operators, a hosted proof gate for teams running coding agents, the SBIR lane for software assurance.

THE $10,000

$4,000 inverter-class hardware and a GPS clock
$3,000 ninety days of the proof gate hosted
$3,000 SBIR preparation and travel

What I need from you:
an introduction at Torus engineering
an introduction at the 309th
a co-founder with power-electronics depth

# THE TRUSTED BASE

- The compiler is written in Mentl and the pinned binary reproduces itself byte for byte; that fixed point is the release.
- The verifier is sound for what it decides and incomplete by design: a claim it cannot decide is reported as pending, never silently passed.
- One false proof is known and documented: a recursion deep enough to exhaust the stack traps although its function says it cannot.
- Rust prevents memory bugs, not effects; Simulink and SCADE verify models and generate C. Neither states "never reaches the network" as an obligation through every callee, proves a function allocation-free, or derives a learning rule from the chain.

# RUNNING ON A CONTROLLER

- The proof is at compile time. The artifact is one WebAssembly module importing WASI and nothing else.
- It runs wherever wasmtime runs, which includes the Linux supervisors that sit beside inverters today; the demo's controller runs that way.
- A native backend is on the roadmap; the pedal in the demo runs the ear natively on a Cortex-M7 today.
- The learner and the predictor keep learning inside a loop proven not to allocate: learning at the edge that cannot stall.

# DURABLE COMPUTATION

A running controller is a value: persisted by copying its image mid-stream and resumed with mentl resume, continuing from the same tick with the same state of charge. The recording is in the repository; the limits are stated there too.

When the grid goes dark, the computation does not.

# THE MESH

## MESH

SIMULATED




Two Stations with GPS time see the **angle** between them, which is the flow on the line between them, and an event's wave arrives at one before the other. FNET has done this from wall outlets since 2004; a fleet that both sees and answers at 10C has not existed. Simulated from the day's shape and the replayed event.